CVIN-SC: The Thesis on One Page — the research side

Nov 9, 2026 · Nikhil Prakash · v0-3 · **Companion to the code-side one-pager of the same date; the two are halves of one picture (see "How this page joins the other one", at the end).**

The thesis

Towards a Minimal Function Specification for Blockchain-Based Self-Sovereign Identity in Connected and Autonomous Systems. Working title; see decision 5. MASc, Electrical and Computer Engineering, UBC. Nikhil Prakash.

Vehicles that talk to each other authenticate under a pseudonym scheme that is unlinkable by design, so no persistent identity history can be attributed to a vehicle. That is a correct design decision with a structural consequence: the registration, authorisation, delegation and revocation operations that cooperative and administrative vehicular services need have no identity layer to run on. This thesis derives the smallest interface such a layer must expose — twelve functions, `IMinimalSSI`, derived from the normative operations of the W3C DID and Verifiable Credential models together with Allen's ten principles of self-sovereign identity — and shows it to be a sufficient and irredundant cover of that requirement set under a published clustering rule (not the unique minimum). It then asks which currently specified blockchain identity standards provide those functions natively, finds that none provides presentation construction (`createPresentation`, the one function that discharges Consent and Minimization), evaluates eight implementations against the interface under a pre-registered protocol, and shows what does and does not transfer when the principal is an autonomous software agent rather than a vehicle.

Research questions and what the evidence currently supports

The thesis carries no hypotheses; it carries three questions, each answered in a stated form at a stated evidence status (verdict · expectation · analysis). The research-side numbers below are the hub's numbers of record (sandbox `cvin-sc`, solc 0.8.24, Hardhat in-process; protocol gas at N = 5 per cell under ruling D-R8).

RQ	Question	Answer as it stands	Status · evidence
RQ1	What operations must an identity interface expose to satisfy the W3C DID/VC normative operations and Allen's principles, and can it be stated as a sufficient and irredundant cover under a declared	Twelve functions in two sets: Set A (eight identifier and key functions) and Set B (four credential functions). Three functions are principle-necessary, five operation-necessary, four conventionally	**analysis**, mechanically checkable (29-operation universe, two elimination tests, graded principle matrix) · §2.3 v2.0, App. M-1, M-2

RQ	Question	Answer as it stands	Status · evidence
	clustering rule?	necessary. The set is closed under the stated lifecycle transitions over the exhibited object set, with a falsification condition; the closure analysis found one gap (early delegate revocation) and amended the specification (F11).	
RQ2	Which blockchain identity standards provide those operations natively, and where one does not, is the shortfall implementation or specification?	Set A is fully native on ERC-1056 only; Set B is native on no EVM standard except as claims or tokens without expiry; `createPresentation` is an anchor on every EVM option and off-ledger on Indy. Eight options pass the same twelve-test suite unmodified. Gas and coverage rank the options differently. A provisional census of 30 ERCs found none specifying holder-side presentation (N = 0, P = 17, X = 13; single rater, not yet official).	**verdict** for ERC-1056 (12/12; eleven of twelve gas values reproduce to the digit across three runs over four and a half months); **protocol gas** on eight options (128/128 cells, N = 5, all repetitions identical); coverage **declared classes, pre-protocol** (D-R7); census **expectation** until executed with a date
RQ3	Does the interface transfer to autonomous software agents, and what does not transfer with it?	The interface transfers; the trust model does not. Five functions transfer exactly, seven modified, none absent; nine non-correspondences named. The specification bounds	**analysis** · §5.2, Appendix I (wrapper at Version 2, no validation rung reached)

RQ	Question	Answer as it stands	Status · evidence
		authority, not intent. A smart-account wrapper is specified, not built.	

Two supporting results worth a paragraph each: an independent on-chain/off-chain placement procedure lands on exactly the Set A / Set B boundary; and "Tier-2 compliance is not a ledger property" — in every scenario the Tier-2 critical path carries only reads and off-chain steps, every write is Tier 3.

The chapters

Under ruling D-R9 (30 October 2026) the hub's five drafted chapters (≈60,200 prose words) and the code side's seven (≈45,000 words by the code side's own count; ≈38,900 at the 31 October text of record, working notes included) become **one seven-chapter thesis** of about 68,000 words, on the hub spine, with the applied system as Chapter 6. The hub side of each chapter and what it rests on:

Chapter	What it argues	What it rests on (research side)
1 Introduction	The accountability gap is structural; the 2020 CVIN architecture assumed the layer this thesis supplies; contributions C1–C14 with their evidence states	Ch. 1 v1-1 (repo); Table 1.1 audited against the corpus
2 Background and related work	SSI principles at definitional strength; W3C standards with a version freeze; V2X and the three-tier latency framework; agent identity	§2.0/2.1/2.2/2.4/2.6 drafts; reservoirs CB-TL / CB-IG; K-15 correction (BSM 10 Hz nominal)
3 The IMinimalSSI specification	The derivation as an irredundant cover under CR-1…CR-3; necessity partition; closure relation and Amendment F11	§2.3 v2.0 — the strongest material in the corpus
4 Methodology	The pre-registered protocol (commit `0a10831`, 15 Oct): N = 5 gas / N = 30 timing, seven void rules, F/P/X rubric, self-audit; the disclosed deviation of the eight-option runs; the eleven ERC-1056 runs that pre-date the protocol, listed by a dated disclosure	§2.5 v1-1 with its D-R7 deviation note; v1-2 amendment drafted

Chapter	What it argues	What it rests on (research side)
	paragraph (v1-2, drafted 5 Nov; trace keys to be confirmed at source before it is appended)	
5 Standards inventory and specification evaluation	Seventeen options in six families; Family E decomposed; the `createPresentation` gap; coverage, protocol gas and deployment gas on eight options	Ch. 3 v0-9; Ch. 4 §4.2–4.4 v1-2; Phase 1 record; D-R7/D-R8
6 Vehicular instantiation	The code side's applied system (W3C layer, V2V, MOBI VID, security, ZK, Steward), labelled by its own measurement regime	Code-side one-pager; spoke import `docs/12_spoke_import/`
7 Discussion and conclusions	The gap as structural; agents; convergence as corroboration; concession-set limitations; RQ answers in the "cover, not minimum" form	Ch. 5 v1-0 §5.1–§5.6

The research side also holds a complete lift-and-assemble draft of the hub thesis in my voice (Take 1, 4 November, ≈27,000 body words) and two further drafting programmes planned and ready to run: a panel rewrite (Take 2) and a provenance-tagged supervisor draft (G-SD1, ten stages) in which every paragraph carries an origin tag so you can see which words are mine.

Decisions I'd like your view on

1. **Admissibility (R-ADM-1 / R-ADM-2).** The eight-option runs of 17–19 October were made after the protocol was pre-registered but not under it; by ruling D-R7 they are admitted with the deviation disclosed, and the gas was re-measured under the protocol (D-R8). Should coverage stay reported as declared classes, or be re-run to rubric verdicts before the defence?
2. **One thesis, one spine (D-R9).** The hub absorbs the applied system as Chapter 6 and keeps RQ1–RQ3 as principal, with the code side's questions as sub-questions (gas, conformance, security, MOBI VID, V2V feasibility). Confirm, or prefer the alternative where the benchmark questions lead?
3. **Contribution count.** C15 (a planned paper) is withdrawn from the table; C16 (revocation latency as a safety budget) and C17 (ordering authority) are proposed; a second "C16" (endorsement-gated credential architecture) from the hybrid-data campaign must be folded or renumbered.

4. **AI assistance.** The drafting programme has used an AI assistant substantively since early 2026 to organise the corpus, structure my results into prose, cross-check numbers against traces and act as a reviewer panel. UBC asks that this be disclosed and approved in writing before drafts are relied on. I would like to send you a one-page disclosure and agreement, with the authorship timeline attached, before any chapter. Throughout these documents, "panel", "lens", "examiner" and "review" name roles performed by the AI assistant under my direction; no one other than me has yet reviewed any of this work.
5. **The title.** The hub title promises a *minimal* specification; the answer the thesis gives is a sufficient and irredundant cover. The code-side title promises a comparison of SSI systems. One title must serve the unified thesis; the candidates are the hub title with "minimal" kept and qualified in §1, or *Measured Trade-offs of Ethereum Identity Standards for Connected Vehicles*. Keep, replace, or something between?

How this page joins the other one

The code-side page describes the applied system by its own five research questions and six hypotheses. Under D-R9 those are not a second thesis; they are sub-questions of this page's RQ2 and RQ3 and one cross-cutting feasibility question answered in Chapter 6. The labels differ on purpose, because the same labels mean different things on the two sides (the code side's RQ1 is gas; this page's RQ1 is the interface):

Code-side question	Where it sits in the unified thesis
RQ1 gas (H1)	RQ2a — what each provision costs, under the protocol (hub) and as native analogues (code side, indicative, different toolchain)
RQ3 W3C conformance (H2)	RQ2b — how far a resolution and issuance layer lifts an on-chain identity to DID Core / VC DM conformance
RQ2 security (H5)	RQ2c — security and recovery properties under the V2X threat model
RQ4 industry / MOBI VID (H4)	RQ2d — realisability of the MOBI VID model across backends
RQ4 real time (H3)	RQ-V — does verification of pre-issued credentials fit the V2V per-message budget (scoped to the verification step)
RQ5 ZK and the Steward (H6)	RQ-V2 (feasibility at registration and session time) and RQ3a (what operating the proofs safely requires of a custodian)

The two sides' gas numbers are never put in one table: the hub measures `IMinimalSSI` functions under a pre-registered protocol on a legacy/paris toolchain; the code side measures native analogues on viaIR/cancun. Both are kept, each in its own table with its toolchain in the caption.